



ASO Author Reflections: Beyond the Cut Surface—Rethinking R1 Margins Through Recurrence and Salvageability

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Past

Microscopically positive (R1) margins have traditionally been viewed as a technical failure after resection of colorectal liver metastases. However, earlier studies yielded conflicting findings regarding whether R1 margin status independently influences survival, particularly in the era of effective systemic therapy and parenchymal-sparing surgery.^{1–3} Moreover, the relationship between margin status, the anatomic pattern of first recurrence, and the opportunity for subsequent liver-directed salvage remained incompletely defined.

Present

In an international multi-institutional cohort of 1060 patients, R1 margin status was associated with worse crude survival outcomes, although the association was weaker after adjustment for primary lymph-node status and hepatic tumor burden and varied according to how missing data were handled.⁴ Importantly, recurrence after R1 resection was less often confined to the liver. Among patients who developed recurrence, isolated liver recurrence occurred in

31.1% of patients after R1 resection versus 47.6% after R0 resection. Repeat hepatectomy was also less common after R1 resection (16.4% vs. 31.4%), with an even larger difference among patients with node-negative primary tumors. These findings suggest that the clinical meaning of an R1 margin extends beyond the cut surface and may reflect a broader oncologic context that influences both recurrence distribution and the feasibility of liver-directed salvage. R0 resection should remain the operative goal; however, an R1 margin may be better interpreted as a context-dependent prognostic marker rather than as a purely technical endpoint.

Future

Future studies should distinguish planned R1 vascular detachment from parenchymal or otherwise unintentional R1 margins and incorporate margin distance, molecular tumor characteristics, and standardized assessment of recurrence location. Further investigation is also needed to determine whether surveillance and multidisciplinary salvage planning informed by margin status and recurrence risk can increase opportunities for repeat liver-directed treatment. Ultimately, integrating technical margin status with tumor biology, disease burden, and anticipated salvageability may provide a more clinically meaningful framework for operative decision-making and post-resection management.

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